



Collapse Behavior and Energy Absorbing Characteristics of 3D-Printed Tubes with Different Infill Pattern Structures: An Experimental Study

Mahmoud M. Awd Allah¹ · Walaa Abdel-Aziem¹ · Marwa A. Abd El-baky¹

Received: 6 March 2023 / Revised: 24 April 2023 / Accepted: 26 April 2023
© The Author(s) 2023

Abstract

This work investigates the impact of infill pattern structure on the crashworthy performance and deformation history of 3D-printed tubes subjected to a quasi-static axial compression load. The proposed tubes were made from polylactic acid (PLA) using a 3D printing technique. Five infill pattern structures were adapted, i.e. circular, square, triangle, zigzag and cross patterns with 50% infill density. The crashworthy analysis was carried out by assessing different indicators, i.e. peak force (F_{ip}), mean crush force (F_m), total absorbed energy (U), specific absorbed energy (SEA), and crushing force efficiency (CFE). It was found that the square infill pattern shows the maximum F_{ip} , F_m , U and SEA with values of (24.38 kN, 20.58 kN, 673.38 kJ, and 26.52) while zigzag infill pattern shows the highest CFE with a value of 0.91. The optimal infill pattern is determined using a Multi-Attribute Decision Making (MADM) method called Complex Proportional Assessment (COPRAS). The COPRAS results showed that the square infill pattern is the most effective energy absorbing structure.

Keywords Crashworthiness · Collapse behavior · Polylactic acid (PLA) · Quasi-static axial load · 3D-printed tubes

Abbreviations

MADM	Multi-attribute decision making
COPRAS	Complex proportional assessment
PLA	Polylactic acid
ABS	Acrylonitrile butadiene styrene
SLA	Stereo lithography
DIW	Direct ink writing
SLS	Selective laser sintering
FDM	Fused deposition modeling
FFF	Fused filament fabrication
FD	Filling density
PA	Polyamide
CCF/PA	Continuous carbon fiber-reinforced polyamide matrix
CGF/PA	Continuous glass fiber-reinforced polyamide matrix
Ψ	Energy absorption effectiveness factor
M_{ac}	Actual specimen mass
M_{ds}	Designed specimen mass
F_{ip}	Peak force
F_m	Mean crush force

U	Absorbed energy
SEA	Specific absorbed energy
CFE	Crushing force efficiency
EAC	Energy absorption capability

1 Introduction

The term “crashworthiness” refers to a material's capacity to protect occupants by absorbing impact energy. An important factor that affects how energy-absorbing devices perform is the type of material utilized in their construction [1]. Energy absorbing of metallic components has been more common in crashworthiness applications during the past two decades [2–4]. To address vital environmental problems, lightweighting has been a key target in the vehicle industry. The mass reduction of the vehicles is valuable for the reduction of fuel consumption and carbon dioxide emissions. To decrease the global weight and fuel consumption of vehicles, more scientists are embracing the usage of energy absorbing of composite components [5–8]. The introduction of energy absorbing of composite components in vehicles not only increases specific energy absorption (SEA) but also reduces noise and vibration, in comparison with metallic components [9].

✉ Marwa A. Abd El-baky
dr.marwa2013@yahoo.com; masalam@eng.zu.edu.eg

¹ Mechanical Design and Production Engineering Department,
Zagazig University, Zagazig 44519, Egypt

Composites can be divided, according to the type of resin, into two categories which are thermosets and thermoplastics. For a number of years, only thermosetting composites could be used. Due to the significant benefits of recyclability, weight reduction, and lower production costs, thermoplastic composites have gained increasing attention in recent years [10, 11]. Different articles on the energy absorption capability (EAC) of thermosetting composites have been published in the literature [12–16].

Due to additive manufacturing, thermoplastic materials like polylactic acid (PLA) and acrylonitrile butadiene styrene (ABS) have gained a lot of interest. Due to various advantages, including high modulus and strength, plastic material is typically employed and developed in commercial services and short-lifecycle applications [17, 18]. There are various traditional techniques for creating thermoplastic cellular structures, however they require too many steps and raise production costs. With the advancement of technology, 3D printing can construct cellular structures more successfully than conventional fabrication techniques. The development and usage of several 3D printing technologies is now underway, including Stereo lithography (SLA), Direct Ink Writing (DIW), Selective Laser Sintering (SLS), and Fused Deposition Modeling (FDM) also referred to Fused Filament Fabrication (FFF) [19, 20]. Thus, numerous researchers have thoroughly examined the mechanical properties of 3D-printed cellular structures with novel core designs for lightweight applications [21–27].

The stiffness and strength of light-weight cellular PLA specimens were investigated by Lubombo and Huneault [28] under edgewise and flatwise uniaxial tensile and flexural loading. The component cells were created utilizing three distinct infill density levels and five different infill pattern types, along with one and three perimeter shells. The part density scaled with the stiffness and strength. At the same density, however depending on the infill patterns, the mechanical response differed greatly. The findings have shown that employing a different sort of infill pattern for the same number of perimeter shells might boost stiffness by up to 2% and strength by up to 82% at the same density. The stiffness and strength at the same density increased by up to 2% and up to 84%, respectively, when more perimeter shells were used for the same infill pattern. Guidelines for part design were developed after examining the scaling factors and rupture modes. Additionally, Ganeshkumar et al. [29] examined the mechanical characteristics of specimens for gyred, rhombus, circular, truncated octahedron, and honeycomb infill architectures (hexagonal). The tensile characteristics of 3D-printed PLA products with regard to their infill pattern were also presented. It was found that in comparison with the alternative infill structures, the hexagonal pattern of infill exhibits superior mechanical properties.

Few studies have investigated the crashworthiness performance of thermoplastic materials; in this regard, the effects of infill pattern, density, and material types on 3D-printed cubes subjected to axial compressive loads under quasi-static circumstances were examined by Quanjin et al. [30]. Both the 70% vol. PLA combined with 30% vol. carbon fiber (PLA/CF) and the 100% PLA were employed. With infill densities of 20, 40, 60, and 80%, four infill pattern structures were produced: triangular, rectilinear, line, and honeycomb. The honeycomb infill design of the 3D-printed PLA cubic structure has the best energy-absorbing qualities when compared to the other three infill patterns. Since they may have better energy-absorbing qualities, the suggested 3D-printed structures, which have a variety of material kinds, infill patterns, and densities, have a considerable potential to replace the present lightweight structures.

In another study, Quanjin et al. [31] examined the impact of five various infill pattern architectures on the energy-absorbing characteristics of single PLA and hybrid (PLA/CF) circular tubes subjected to quasi-static axial load. Utilizing additive manufacturing and filament winding, tubes were created. Adapted patterns included regular triangular, square, hexagonal, and tetrahedral shapes. It was discovered that the infill pattern structure significantly affects the qualities that absorb energy. Ebrahimi et al. [32], experimentally and numerically, investigated the in-plane (EAC) and uniaxial compressive reactions of new structures fabricated by additive manufacturing techniques. Results revealed that unit-cell arrangement has a pronounced impact on the mechanical properties and (EAC). Layer heights and infill patterns were investigated by Tsouknidas et al. [33] in relation to the ability of 3D-printed PLA cylinders created using (FDM) technology to withstand impacts. The layer heights and infill patterns were found to be less significant and insignificant, respectively, according to the results. On the other hand, higher (EAC) was correlated with higher infill densities.

Under axial impact loading, Zahran et al. [34] examined the (EAC) of single-cell, conventional multi-cell and three-dimensional multi-cell (3D-multi-cell) structures. The numerical results confirmed that 3D-multi-cell structure can significantly outperform circular tube in terms of (SEA) up to 40%, (CFE) up to 42.5%, and the energy absorption effectiveness factor (Ψ) up to 30.7%. Additionally, it may significantly increase the SEA up to 20%, the (CFE) up to 36.8%, and the (Ψ) up to 20% when compared to those of typical multi-cell structures. Additionally, it was discovered that (EAC) and the (CFE) respond differently depending on the number of layers and angles of orientation. Quasi-static and dynamic performances of 3D-printed hollow profiles made from continuous carbon (CCF/PA) and glass (CGF/PA) fibers-reinforced polyamide matrix, in axial and radial directions, were studied by Morales et al. [35]. Results

indicated that CCF/PA profiles indicated SEA values bigger than CGF/PAs under radial quasi-static circumstances. However, CGF/PA profile's radial impact performance was better because the material had a strain-rate dependency. The radial SEA values for the steered glass fibers were lower than their axial loading values. SEA outcomes are encouraging. They exceeded any values found in the literature by a factor of two to three. The results suggest that impact laden hollow profile applications could use concentrically printed CGF/PA.

Wang et al. [36] explored the crashworthiness features of multi-cell filled thin-walled structures produced by carbon fiber-reinforced PA using (FDM) process. All studied structures had identical exterior hexagonal shape but with circular, hexagonal and triangular interiors with different densities. The crushing performance of the constructions were investigated via dynamic and quasi-static tests. According to the experimental findings, (SEA) increased as filling density (FD) increased. When comparing specimens with high (FD) to those with low density, it displayed an increase of 127%. The triangular construction with medium (FD) displayed the maximum value of 85.6% when the (CFE) increased initially, and then declined. (SEA) of the filled circular structures was higher than that of the filled hexagonal and triangular ones. Additionally, all constructions' (SEA) and (CFE) at great velocity reduced by 59 and 39%, respectively, as compared to quasi-static load. The failure mode of thin-walled structures changed from ductile damage mode to brittle fracture mode as a result of the composite materials' strain rate influence.

Under quasi-static and dynamic loads, Wang et al. [37] investigated the progressive collapse behaviors and mechanisms of 3D-printed thin-walled composite structures fabricated by (FDM) process using PA and fiber-reinforced PA-based composites. Circular, triangular, quadrangular, and hexagonal cross-sections were the four configuration types studied. It was discovered that under quasi-static compression, thin-walled composite structures displayed a gradual, stable, and regular plastic deformation mode. The thin-walled constructions displayed a gradual brittle crack-to-fracture collapse mode under low-velocity impact conditions. It was found that (F_{ip}) increased while (SEA) dropped as the loading velocity was increased. Under both quasi-static compression and dynamic impact circumstances, the energy absorption performance of the hexagonal constructions made of carbon fiber-reinforced material was at its best.

More research efforts should be intensified towards the effect of the infill pattern structure on crashworthy 3D-printed components for modern requests. The current work suggests new infill patterns. The goal of the research is to create a structure that can absorb the required amount of energy while achieving a lower peak load (F_{ip}). The lowering of (F_{ip}) is a crucial requirement because it directly

affects the structure's and/or the passengers' safety. The greater force will result in a greater deceleration, which could cause harm to people and structures. Additionally, the crushing should be done in a controlled manner. This study is an initial trial to investigate the effect of different infill pattern structures on the crashworthiness performance and deformation history of 3D-printed polylactic acid (PLA) cylinders subjected to quasi-static axial load. Five infill pattern structures, i.e. circular, square, triangle, zigzag and cross patterns, were adapted at 50% infill density. In addition, Complex Proportional Assessment (COPRAS) has been used to find the optimal infill pattern structure.

2 Materials and Methods

2.1 Materials

A PLA filament spool was supplied by TP Plastics Co. Ltd. (Egypt) to explore the influence of infill structures on the crushing performance of 3D-printed tubes. Table 1 demonstrates the physical and mechanical properties of PLA filament as provided by the vendor. PLA filament was chosen for this study due to its low cost, biodegradability, commercial availability, good mechanical and thermal properties, and low shrinkage and thermoforming dimensional stability characteristics. To reduce the influence of moisture conditions on 3D-printed tubes, PLA filament was stored in a vacuum bag before the fabrication process.

Table 1 Physical and mechanical properties of PLA as provides by the vendor

Property	PLA
Color	Green
Cost (£/kg)	32
Density, (g/cm ³)	1.24
Content (vol.)	100% PLA
Melting temperature (°C)	210
Young's modulus (GPa)	3.5
Tensile strength (MPa)	60
Shear modulus (MPa)	1287
Poisson's ratio	0.36
Filament diameter (mm)	1.75 ± 0.05
Filament weight (kg)	1
Standard tolerance (mm)	±0.05
Shore hardness	39 D
Extruder temperature (°C)	180–230
Bed temperature (°C)	20–60

2.2 Specimens Fabrication Method

FDM often referred to as material extrusion, is now the most popular additive manufacturing technology available. FDM divides the model into thin layers where polymer filament is deposited to sketch the shape and fill the internal area layer by layer. Its success is primarily down to its affordable printing method, low cost, and how simple it is to find parts Jin et al. [38]. Before printing, the filaments were sealed and stored at 20 °C. A commercial 3D printer Prusa® i3 MK3 based on FDM technique was used to manufacture the PLA proposed tubes. PLA filaments were fed then heated to melt in the heating device and then extruded to the printing platform by the nozzle. The nozzle extrudes a thin filament of molten plastic based on the contour of the current layer, creating the final component layer by layer in accordance with a CAD file. The build platform is adjusted in the z-direction to control the layer thickness. The printing speed was 50 mm/s. The temperatures of the heating nozzle and printing platform were 210 °C and 50 °C, respectively.

Test specimens were fabricated through FDM using the Prusa® i3 MK3 printer provided by Prusa Research Company, see Fig. 1. It has a number of parts, including the LCD screen, control buttons, heated bed, extruder, frame structure, and type of filament. The outside temperature was kept at or near 25 °C during the manufacturing process to avoid the ambient temperature. Table 2 lists the constant process variables for creating the test specimens. As stated by Dawoud et al. [39] the raster angle of +45°/−45° for the FDM parts can result in higher mechanical and fracture

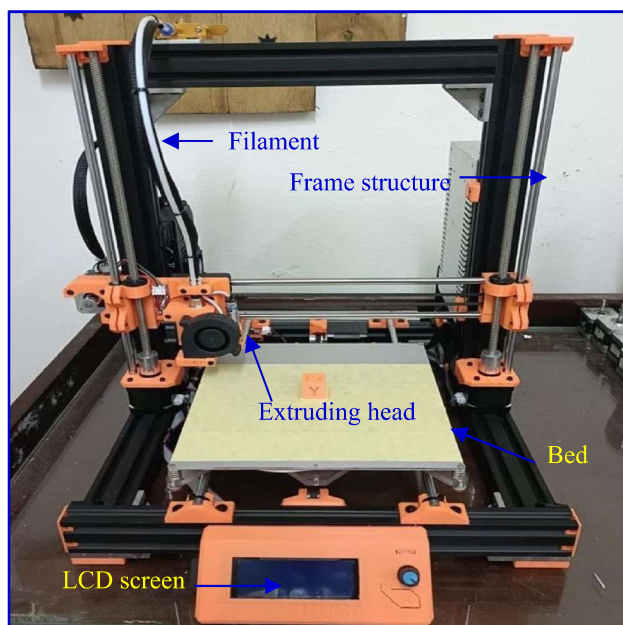


Fig. 1 3D printing machine used in this work

Table 2 Operating parameters for PLA specimens

Operating parameters	PLA
Nozzle diameter (mm)	0.4
Filament diameter (mm)	1.75 ± 0.05
Extrusion width (mm)	0.45
Extruder temperature (°C)	210
Bed temperature (°C)	53
First layer height (mm)	0.20
Base layer type	Brim
Flow rate (%)	100
Brim size (mm)	3
Build direction	Flat, printed through the thickness
Raster angle (°)	+45°/−45°
Max volumetric speed (mm/s)	15

resistance; therefore, in the present study this raster orientation was selected for the fabrication of the specimens.

The constructed specimens were subjected to tensile and quasi-static compression tests. In order to simulate continuous material as precisely as possible for tensile specimens, a notional infill density of 100% was established for all test specimens by Marşavina et al. [40]. Tensile test specimen dimensions according to ASTM D638-V are shown in Fig. 2.

For the quasi-static experiment, five types of infill patterns were adapted in this study, i.e., circular, square, triangle, zigzag and cross patterns. The infill patterns were selected from the infill function of Prusa software. The wall thickness of the cylinders was set as 0.5 mm, and it was measured using a Vernier caliper. For all test specimens, a nominal infill density of 50% was established. SolidWorks® was used to create the test specimen geometry, which was then exported as a Stereo lithography (STL) file for 3D printing software. The geometrical description for the 3D-printed PLA cylinders is displayed in Fig. 3. The details about 3D-printed tubes are summarized in Table 3.

2.3 Test Procedure

Tensile and quasi-static tests were carried out on a universal testing machine (type: Jinan WDW 100 kN) at room temperature. Three identical specimens were tested in each case and the average was delivered.

2.3.1 Tensile Test

A tensile test was carried out according to ASTM Standard 638-14 (Type V specimens). Specimens were subjected to a crosshead speed of 1 mm/min until failure of the specimen occurred. A tensile test produced a load–displacement curve,

Fig. 2 Tensile test specimen dimensions according to ASTM D638-V. All dimensions are in mm

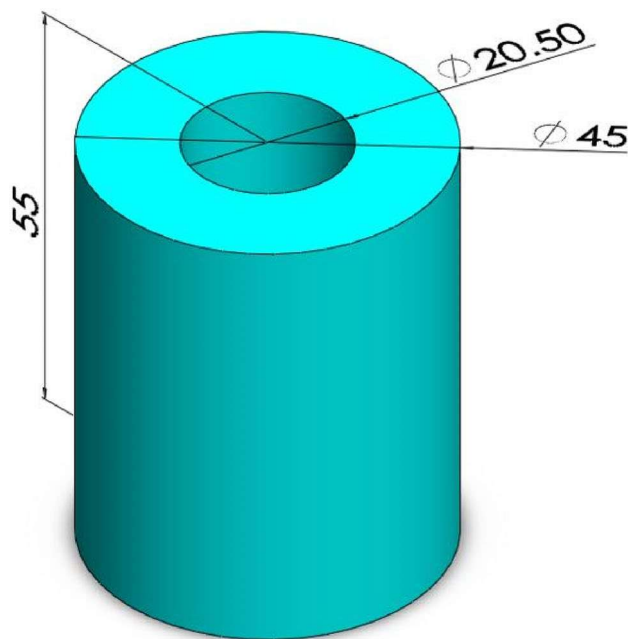
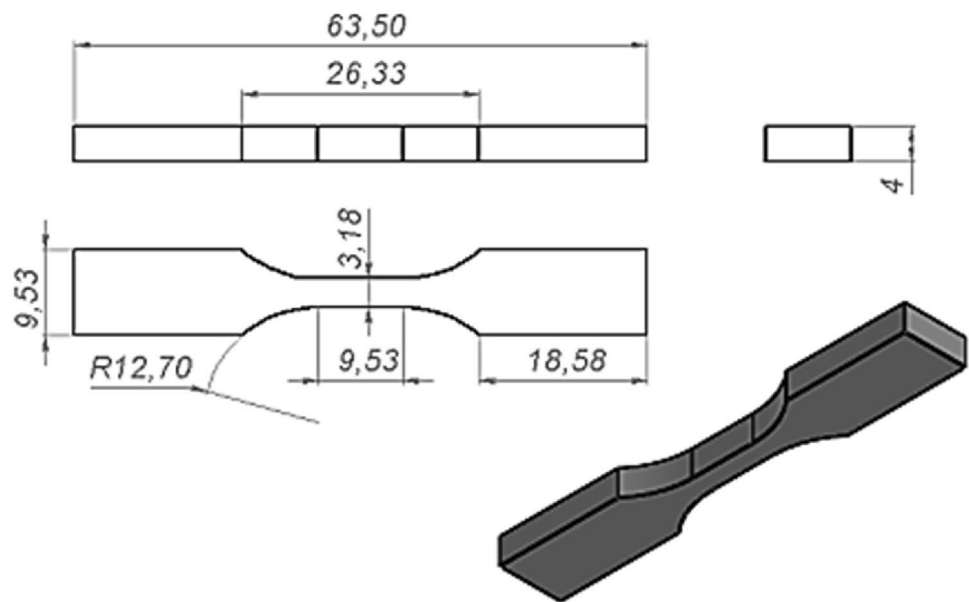


Fig. 3 Geometrical description of 3D-printed cylinder

which was analyzed and then transformed into a stress–strain curve. The tensile strength, Young's modulus, and strain to failure were then calculated.

2.3.2 Quasi Static Test

The quasi-static test is straightforward, manageable, and reasonably priced so it was adapted in the present study [41, 42]. Prior to the start of the test, each 3D-printed tube was placed between two flat steel plates that were

parallel to each other. A compressive force was applied to the specimen. The crushing behavior of a 3D-printed tube was captured on a camera. The rate of crushing for all specimens was 2 mm/min [16, 43–45]. The obtained load–displacement profiles can be used to measure the crashworthy performance of 3D-printed tubes. The crushing critical indicators were calculated [14].

2.4 Complex Proportional Assessment (COPRAS)

The best infill pattern structure among several options is typically chosen using multi-attribute decision making (MADM) methods like the, Analytical Hierarchy Process (AHP), Method of Order Preference Similarity to the Ideal Solution (TOPSIS), Preference Ranking Organization Method for Enrichment Evaluations (PROMETHEE), and Complex Proportional Assessment (COPRAS). Due to its ease, COPRAS was employed in the current search to identify the infill pattern structure that had the optimal combination of energy absorption indicators. The tubes examined in this study were taken into account as design alternatives inside the COPRAS framework, where they are rated according to their relative significance in relation to the performance criteria for energy absorption. Below is a description of how COPRAS was used in its entirety to build an energy-absorbing structure [46, 47].

Step 1: Progress a preliminary decision matrix X.

The 1st stage in applying COPRAS is to create a preliminary decision matrix X charting the contenders, i.e. the tubes, to their attributes, i.e. energy absorption indicators. The matrix X can be stated as:

Table 3 Details about the fabricated 3D-printed tubes with different infill pattern structures

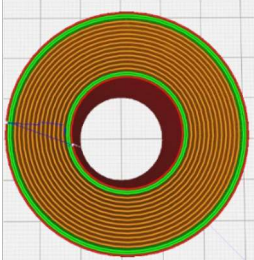
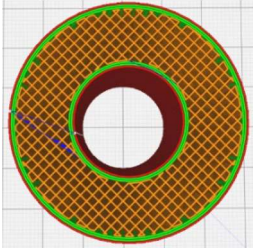
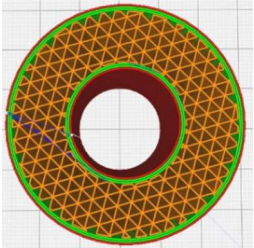
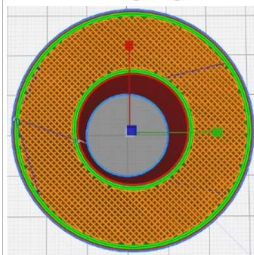
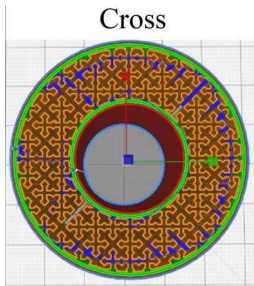
Infill patterns tested	Layer thickness (mm)	Time (min)	Infill density (%)	Wall thickness in green (mm)	Tube thickness (mm)	Actual specimen mass M_{ac} (g)	Designed specimen mass M_{ds} (g)
Circular	0.28	116	50	1.2	$\frac{D_o - D_i}{2} = \frac{25 - 20.5}{2} = 12.25$	41.38	50
 Circular							
Square	0.28	126	50	1.2	12.25	42.64	53
 Square							
Triangle	0.28	126	50	1.2	12.25	43.48	53
 Triangle							
Zigzag	0.28	125	50	1.2	12.25	43.15	53
 Zigzag							

Table 3 (continued)

Infill patterns tested	Layer thickness (mm)	Time (min)	Infill density (%)	Wall thickness in green (mm)	Tube thickness (mm)	Actual specimen mass M_{ac} (g)	Designed specimen mass M_{ds} (g)
Cross	0.28	148	50	1.2	12.25	38.99	47



$$X = [x_{ij}]_{mn} = \begin{bmatrix} x_{11} & \dots & x_{1j} & \dots & x_{1n} \\ \vdots & & \vdots & & \vdots \\ x_{i1} & \dots & x_{ij} & \dots & x_{in} \\ \vdots & & \vdots & & \vdots \\ x_{m1} & \dots & x_{mj} & \dots & x_{mn} \end{bmatrix}; i = 1, \dots, m, j = 1, \dots, n \quad (1)$$

where m and n are, respectively, the contenders number and the indicators number. Thus, x_{ij} signifies the performance of the i_{th} tube in terms of the j_{th} indicator.

Step 2: Create a normalized decision matrix R :

Meanwhile the changed design indicators have dissimilar units, they cannot be associated directly to each other and this makes the choice harder. So, the matrix X must be transformed into a no dimensional one to make the design indicators comparable. The no dimensional, normalized, decision matrix R is expressed as:

$$R = [r_{ij}]_{mn} = \frac{x_{ij}}{\sum_{i=1}^m x_{ij}} \quad (2)$$

where r_{ij} denotes the normalized j_{th} indicator for i_{th} design.

Step 3: Calculate the separate weightage of each attribute (w_j). w_j for each attribute can be determined as follows:

- Each two indicators need to be compared at a time. This will result in a total number of assessments equal to $N = (n(n-1)/2)$. When the compared indicators are of dissimilar significance for the selection process, a score of 3 should be given to the more important indicator while a score of 1 should be set to the less significant indicator. If the compared indicators are equally significant for the choice process, then a score of 2 can be given to both of them.
- The total score for each indicator can be obtained as:

$$W_j = \sum_{i=1}^N w_{ij} \quad (3)$$

- The weightage of the j_{th} indicator can be obtained by dividing the total score of each indicator w_j by the global total score H :

$$w_j = \frac{W_j}{H} = \frac{W_j}{\sum_{j=1}^N W_j} \quad (4)$$

Step 4: Define the weighted normalized decision matrix D :

D matrix can be gotten by multiplying the normalized matrix R by the individual weightage of each indicator and it can be written as shown:

$$D = [y_{ij}]_{mn} = r_{ij} \times w_j \quad (5)$$

where y_{ij} is the weighted normalized j_{th} indicator for i_{th} design.

Step 5: Discover the summation of the positive and negative indicators for each design alternative.

The matrix D holds together positive and negative indicators denoted, respectively, as $+y_{ij}$ and $-y_{ij}$. The respectable design contender is the one that has a greater value for the positive indicator and a smaller value for negative indicator. Therefore, for each design alternative the summation of the positive (S_{+i}) and negative (S_{-i}) indicators would be found as follow:

$$S_{+i} = \sum_{j=1}^n +y_{ij} \quad i = 1, \dots, m \quad (6)$$

$$S_{-i} = \sum_{j=1}^n -y_{ij} \quad i = 1, \dots, m \quad (7)$$

Step 6: Compute the relative significance (Q_i) and the quantitative utility (U_i) for each design alternative.

In order to characterize the significance of the design contenders in relation to the selection criteria, the main techniques obtained as a result of applying COPRAS are Q_i and U_i . These formulae can be used to create them:

$$Q_i = S_{+i} + \frac{\sum_{i=1}^m S_{-i}}{S_{-i} \sum_{i=1}^m \frac{1}{S_{-i}}} \quad (8)$$

$$U_i = \frac{Q_i}{Q_{\max}} \times 100 \quad (9)$$

The design option with the highest Q_i and U_i values is the proper one. Anywhere the U_i value is used to rank the candidates, the option with the highest U_i is placed first and the option with the lowest U_i is placed last.

3 Results and Discussions

3.1 Tensile Characteristics

The typical tensile stress–strain curves of 3D-printed PLA specimens are shown in Fig. 4. As can be observed, the sample first behaved in a linear elastic manner up to a stress of around 40 MPa, then transitioned into a non-linear zone until the maximum stress was attained. After the maximum stress peak, there was a stress reduction with rising strain till failure. This is consistent with that recorded by Agaliotis et al. [48].

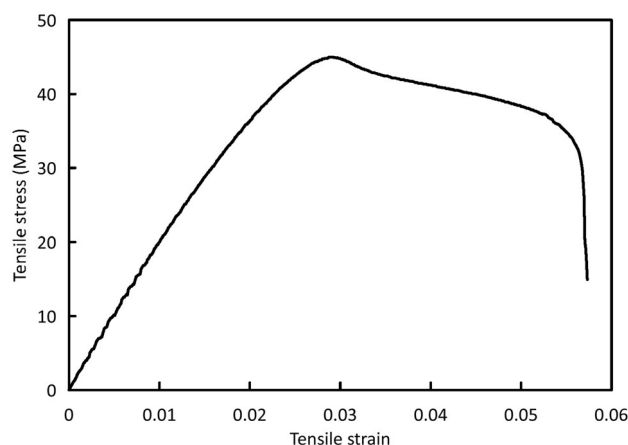


Fig. 4 Tensile stress–strain curve for PLA

Table 4 displays the printed material's tensile Young's modulus, maximum stress, and strain to failure. Young's modulus of 3D-printed PLA specimens was found to be 2.01 GPa, significantly less than the manufacturer's stated value (3.5 GPa). According to Carrasco et al. [49] and Liao et al. [50] greater porosity caused by the printing process, void development between printed layers and beads, and the decrease in molecular weight brought on by the extrusion and printing processes all contribute to this drop in mechanical characteristics. Additionally, the mechanical performance of the material may have been impacted by debonding between adjacent deposited beads [39] and variations in the degree of crystallinity during the FDM process [51]. As shown in Table 4, the voids generated between the layers are the reason why the maximum stress of the printed PLA (40 MPa) is lower than the manufacturer's quoted value (60 MPa) [52].

3.2 Quasi-Static Characteristics

3.2.1 Load–Displacement Profile and Deformation History

Load–displacement profiles and the deformation histories obtained from the quasi-static axial compression tests for 3D-printed tubes are shown in Figs. 5, 6, 7, 8, 9. Outcomes, existing in Figs. 5, 6, 7, 8, 9, are for the most illustrative sample for each infill pattern structure. Generally, it is clear

Table 4 Tensile properties of PLA obtained from tensile test

Property	Value
Young's modulus (GPa)	2010 ±60
Failure strain	0.057
Tensile strength (MPa)	44.97

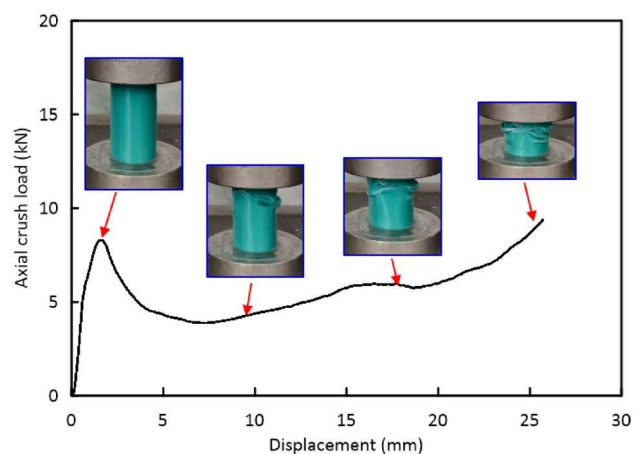


Fig. 5 Load–displacement and deformation history for circular infill pattern tube

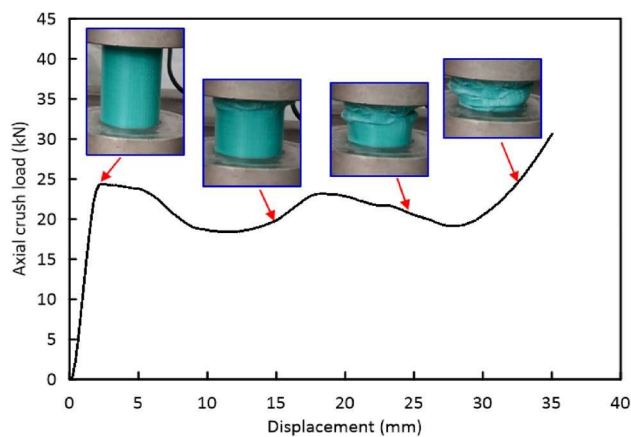


Fig. 6 Load–displacement and deformation history for square infill pattern tube

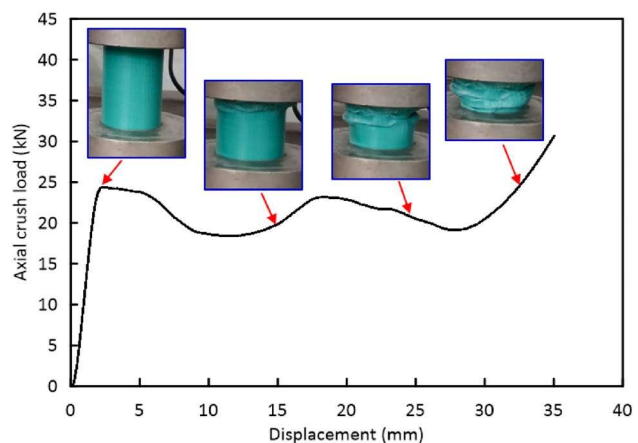


Fig. 9 Load–displacement and deformation history for cross infill pattern tube

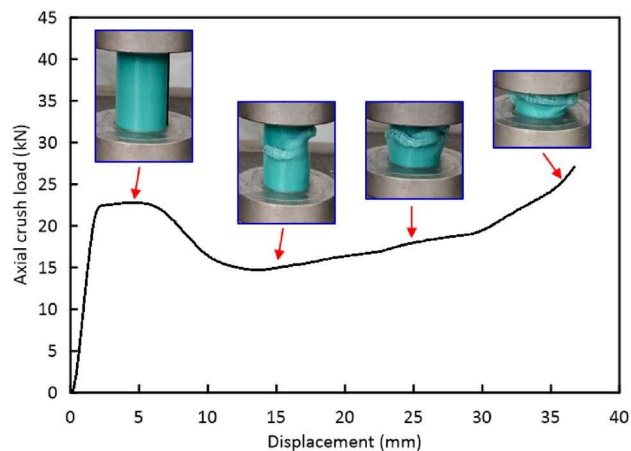


Fig. 7 Load–displacement and deformation history for triangle infill pattern tube

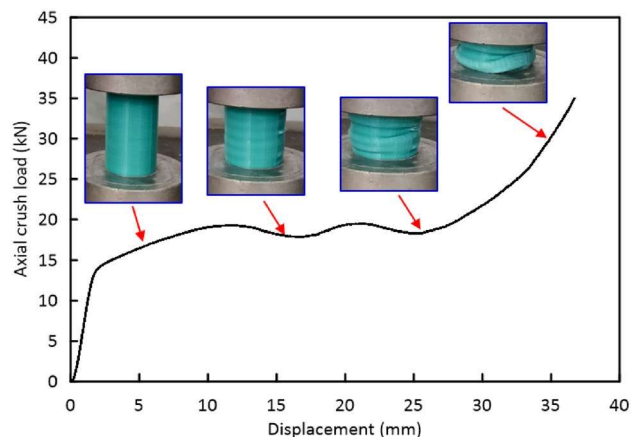


Fig. 8 Load–displacement and deformation history for zigzag infill pattern tube

that all tubes rapidly come up to the initial first peak loads of 8.30, 24.38, 22.78, 19.28, and 20.19 kN at displacements, respectively, of 1.63, 2.38, 4.10, 11.76 and 1.92 mm for circular, square, triangle, zigzag and cross infill patterns. After that all tubes enter the post-crushing zone and the load oscillates around the mean crush force. Tubes go into densification zone at (7.91 kN, 24.05 mm), (25.01 kN, 32.75 mm), (22.28 kN, 33.12 mm), (21.75 kN, 30.04 mm), and (18.40 kN, 36.11 mm) for, respectively, circular, square, triangle, zigzag and cross infill patterns. When tubes enter the densification stages, the crush load rises rapidly due to the full damage of the specimens. Thus, they cannot survive extra loads. At this instant, the table of the machine holds the specimen and resists the force which rapidly increases. This result agrees with that reached by Awd Allah et al. [5], Alshahrani et al. [53], and Abdewi et al. [54].

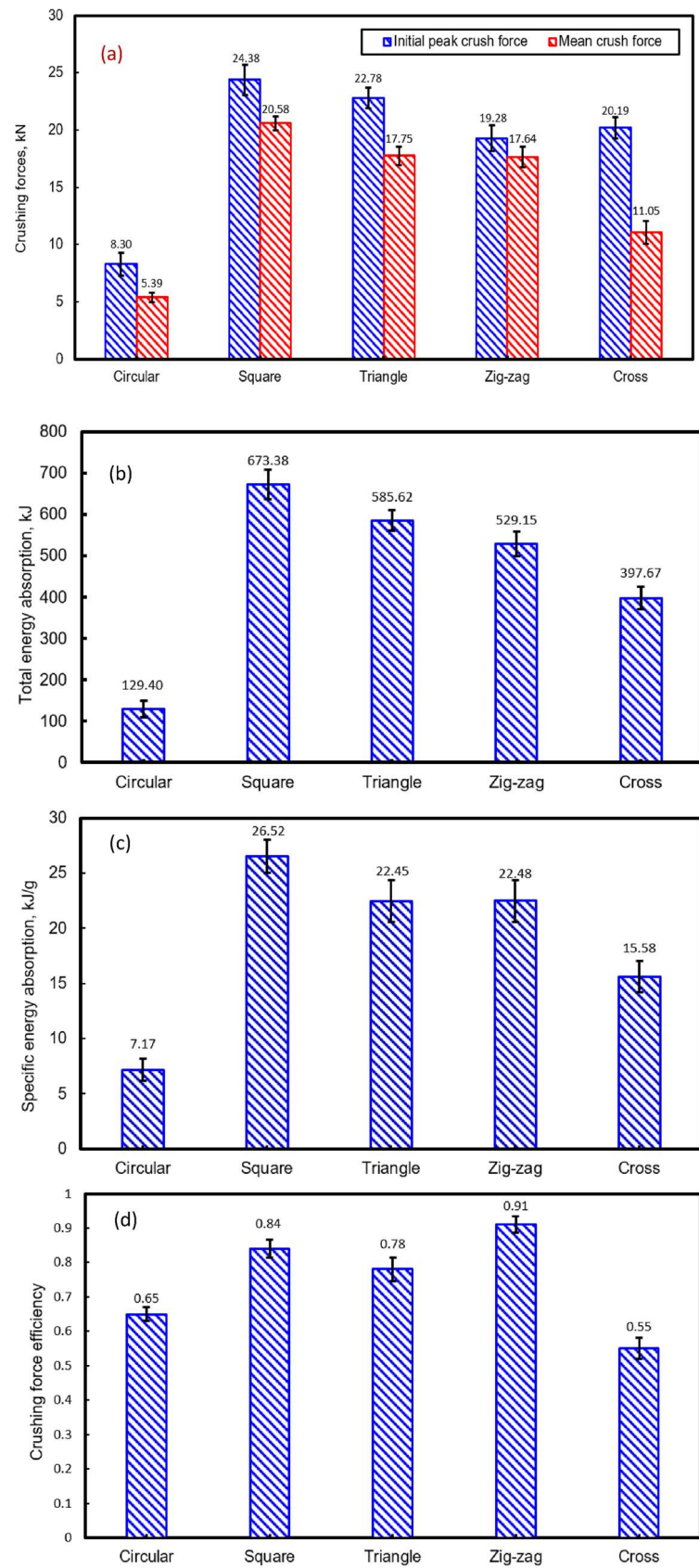
3.2.2 Crashworthiness Indicators

The average values of the crashworthiness indicators for 3D-printed tubes with different infill patterns obtained from quasi-static tests are presented in Fig. 10.

a) Initial peak load (F_{ip}) and mean crushing load (F_m)

As reported by Awd Allah et al. [16] and Guler et al. [55] smallest F_{ip} is desirable to avoid the alteration of crash energy from the absorber unit to the main truck body. As shown in Fig. 10(a), the lowest (F_{ip}) and (F_m) were documented for circular infill pattern tube with a value of 8.30 and 5.39 kN, respectively. Whilst the maximum (F_{ip}) and (F_m) were noted for square infill pattern tube with a value of, respectively, 24.38 and 20.58 kN. (F_{ip}) and (F_m) of square infill pattern tube are, respectively, 2.94 and 3.82 times of those of circular infill pattern tube. (F_{ip}) for triangle, zigzag

Fig. 10 Crashworthiness indicators for PLA 3D-printed tubes



and cross infill pattern tubes are 22.78, 19.28 and 20.19 kN with increasing percent's of 175, 132 and 1.43%, respectively, over circular infill pattern tube. Also, (F_m) were 17.75, 17.64 and 11.05 kN with enhancement percent of 229, 227 and 105%, respectively, for triangle, zigzag and cross infill pattern tubes compared with circular infill pattern.

b) Total absorbed energy (U)

As represented in Fig. 10(b), the lowest (U) value was detected for circular infill pattern tube with a value of about 129.40. The maximum (U) was recorded for square infill pattern tube with a value of about 673.38 J, with an enhancement percent of 420% over circular infill pattern tube. Also, (U) for triangle, zigzag and cross infill pattern tubes are 585.62, 529.15 and 397.67 J with enhancement percent's of 353, 309 and 207%, respectively, over circular infill pattern tube.

c) Specific energy absorption (SEA)

It is clear from Fig. 10(c) that infill pattern shape has a great influence on SEA. Figure 10(c) shows that, the lowest SEA value was noticed for circular infill pattern tube with a value of about 7.17 J/g, but the maximum SEA was documented for square infill pattern tube with a value of about 26.52 J/g, with an improvement of 270% over circular infill pattern. Similarly, SEA for triangle, zigzag and cross infill pattern tubes are, respectively, 22.45, 22.48 and 15.28 J/g with enhancement percent's of 213, 214 and 113% over circular infill pattern tube.

d) Crushing force efficiency (CFE)

High CFE is desirable as it means high crashworthy performance of the structure [56, 57]. As denoted in Fig. 10(d), the lowest CFE was recorded for cross infill pattern tube with a value of about 0.55, but the maximum CFE was documented for zigzag followed by square infill pattern tubes with a value of about 0.91 and 0.84, with enhancement percent of 65.45 and 52.73% over cross infill pattern. Likewise,

CFE for circular, square and triangle infill pattern tubes are 0.65, 0.84 and 0.78 with improvement percent's of 18.18, 52.73 and 41.82% over cross infill pattern tube, respectively.

3.3 Failure Mechanisms

Crashworthy components are mostly designed to absorb maximum amount of crush energy. The actual mechanisms and sequence of failure are highly dependent on the geometry of the structure, infill density, testing speed and failure initiators such as cutouts, slots, and holes, all of which can be appropriately planned to progress high energy absorbing mechanisms [53, 58, 59]. Snaps of top views for the crushed tubes are shown in Fig. 11. Failure modes can be classified as follows:

Mode I: Internal and external folds caused undesirable performance, i.e. low total and specific absorbed energy as shown in the circular infill pattern.

Mode II: Progressive folds started from the top end of the tubes causing unbalanced distortion as shown in square and cross infill patterns. This mode caused an enhancement in the crashworthiness performance.

Mode III: Small uniform symmetric wrinkles were performed in the tube wall as shown in the zigzag infill pattern.

Mode IV: Mid-length collapse including a form of catastrophic failure at the center of the tube as shown in the triangle infill pattern, resulting in an adequate performance.

3.4 Determining the Optimal Infill Pattern

The initial COPRAS decision-making matrix, illustrated in Table 5, including five infill patterns, i.e. circular, square, triangle, zigzag, and cross patterns, as well as five attributes that detail their crashworthiness indicators, including F_{ip} , F_m , U, SEA and CFE. Furthermore, The normalized non-dimensional COPRAS decision matrix (R) of F_{ip} , F_m , U, SEA and CFE were calculated according to the procedure explained Previously in Sect. 2.4 and detailed in Table 6. After that, the individual weightage of F_{ip} , F_m , U, SEA and CFE were determined. All crashworthiness indicators were considered equal in importance for



Fig. 11 Failure signs of 3D-printed tubes with different infill patterns

Table 5 Initial COPRAS matrix (X)

Infill patterns	Attributes				
	F_{ip} (kN)	F_m (kN)	U(J)	SEA(J/g)	CFE
Circular	8.30	5.39	129.4	7.17	0.65
Square	24.38	20.58	673.38	26.52	0.84
Triangle	22.78	17.75	585.62	22.45	0.78
Zigzag	19.28	17.64	529.15	22.48	0.91
Cross	20.19	11.05	397.67	15.58	0.55

Table 6 The normalized COPRAS decision matrix (R)

Infill patterns	Attributes				
	F_{ip}	F_m	U	SEA	CFE
Circular	0.087433	0.074437	0.055891	0.076115	0.174263
Square	0.256821	0.284215	0.290849	0.281529	0.225201
Triangle	0.239966	0.245132	0.252944	0.238323	0.209115
Zigzag	0.203097	0.243613	0.228553	0.238641	0.243968
Cross	0.212683	0.152603	0.171763	0.165393	0.147453

Table 7 Individual weighting for each crashworthiness indicators

	$N = \frac{n(n-1)}{2} = \frac{5(5-1)}{2} = 10$				
	F_{ip}	F_m	U	SEA	CFE
W_{ij}	2	2	—	—	—
	2	—	2	—	—
	2	—	—	2	—
	2	—	—	—	2
	—	2	2	—	—
	—	2	—	2	—
	—	2	—	—	2
	—	—	2	2	—
	—	—	2	—	2
W_j	8	8	8	8	8
G	40	—	—	—	—
w_j	0.20	0.20	0.20	0.20	0.20

the selection process and therefore they were giving a score of 2 when compared with each other. The process of classifying the individual weightage is listed in Table 7. Turning to the next step of calculated the weighted normalized decision matrix (D) that presented in Table 8. To define S_{+1} and S_{-1} , F_m , U, SEA and CFE were considered as valuable responses, while F_{ip} was established as non-valuable attribute. Subsequent to that, the values of Q_i and U_i for each infill pattern were calculated and existing along with the rank of the infill pattern in Table 9. Based on the COPRAS results, square and circular infill pattern were ranked first and last, respectively.

Table 8 Weighted normalized decision matrix (D)

Infill patterns	Attributes				
	F_{ip}	F_m	U	SEA	CFE
Circular	0.017487	0.014887	0.011178	0.015223	0.034853
Square	0.051364	0.056843	0.058170	0.056306	0.045040
Triangle	0.047993	0.049026	0.050589	0.047665	0.041823
Zigzag	0.040619	0.048723	0.045711	0.047728	0.048794
Cross	0.042537	0.030521	0.034353	0.033079	0.029491

Table 9 COPRAS results

Infill patterns	Q_i	U_i	Rank
Circular	0.773418	63.630137	5
Square	1.21549	100	1
Triangle	1.0886	89.560589	3
Zigzag	1.123837	92.459584	2
Cross	0.798654	65.706341	4

3.5 Applications

The innovative proposed tubes with different patterns can be used as energy-absorbing elements in the front of railway or automotive constructions, such as anti-impact rods or a crash box, as well as in an aircraft fuselage. For high-performance applications and safety equipment in significant transportation industries including the aerospace, automotive, and marine industry, hybrid composites into energy absorbing devices or crash boxes can be built, as shown in Fig. 12.

4 Future Work

The current study can be broadened to explore the effect of other crucial issues on the crashworthiness performance of 3D tubular structures as follows: -

- Studying the effects of loading rate, temperature, and parametric conditions i.e., slenderness ratio, wall thickness, and cross-sectional shape.
- Optimization of the abovementioned parameters by finite element analysis.
- Verifying the numerical approximation and analytical solution with measurements. It is recommended to use numerical models to reduce cost and to expedite the understanding of the structure's response.

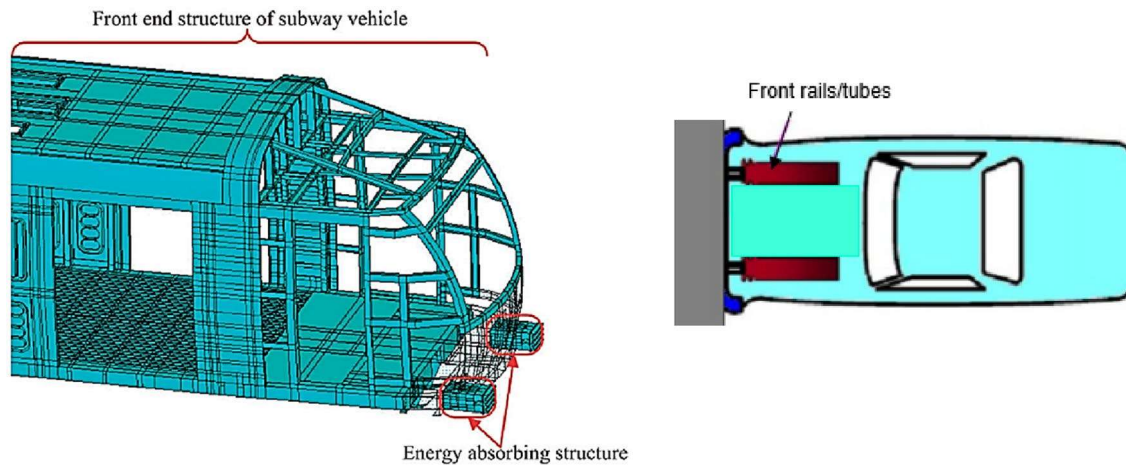


Fig. 12 Applications of the proposed tubes

5 Conclusions

The current work investigates the effect of the infill of pattern structure on the crashworthiness performance and deformation history of PLA tubes. Five infill pattern structures were adapted, i.e. circular, square, triangle, zigzag and cross patterns at 50% infill density. The proposed tubes were fabricated via 3D printing technique and tested under quasi-static axial compression. Then, optimal infill pattern is assessed using a Multi-Attribute Decision Making (MADM) method called Complex Proportional Assessment (COPRAS). The following observations have been recognized:

Infill pattern structure has a significant effect on the crashworthiness performance of PLA 3D-printed tubes. The highest (F_{ip}) value was noted for square followed by triangle infill patterns with values of 24.38 and 22.78, respectively. While the smallest (F_{ip}) was recorded for circular infill pattern with a value of 8.30 kN. The highest U was recorded for square followed by triangle infill patterns with a value of, respectively, 673.38 and 585.62 J, while the lowest U was reported for circular infill pattern with a value of 129.40 J. The highest SEA was recorded for square followed by zigzag infill pattern with a value of 26.52 and 22.48 kJ/g, respectively. The lowest SEA was recorded for circular infill pattern with a value of 7.17 kJ/g. The highest CFE was documented for zigzag followed by square infill patterns with a value of 0.91 and 0.84, respectively. The lowest CFE was noted for cross infill pattern, with a value of 0.55. The failure mechanism of PLA 3D-printed tubes can be changed and affected by the infill pattern structure. A square infill pattern is recommended for crashworthy structures whilst circular infill pattern is not recommended for energy absorbing

components in automotive applications based on the COPRAS results.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB).

Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of Interest No potential conflict of interest was reported by the authors.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. C.W. Isaac, C. Ezekwem, Compos. Struct. (2021). <https://doi.org/10.1016/j.compstruct.2020.113081>
2. J. Marzbanrad, A. Abdollahpoor, B. Mashadi, Int. J. Crashworthiness **14**, 591 (2009)
3. Z. Wang, X. Jin, Q. Li, G. Sun, Int. J. Mech. Sci. **171**, 105380 (2020)
4. M. Kathiresan, Thin-Walled Struct. **154**, 106793 (2020)
5. M.M. Awd Allah, M.A. Abd El-baky, M.A. Hassan, A. Shaker, Fib. Polym. **23**, 3268 (2022)

6. M.A.A. El-Baky, M.M.A. Allah, M. Kamel, W. Abd-Elaziem, *Sci. Rep.* **12**, 21097 (2022)
7. M.A.A. El-Baky, M.M.A. Allah, M. Kamel, W. Abd-Elaziem, *Sci. Rep.* **12**, 21101 (2022)
8. H. Alshahrani, T.A. Sebaey, M.M. Awd Allah, M.A. Abd El-baky, *J. Compos. Mater.* **57**(7), 1315–1330 (2023)
9. H. Zarei, M. Kröger, H. Albertsen, *Compos. Struct.* **85**, 245 (2008)
10. S. Boria, A. Scattina, G. Belingardi, *Compos. Struct.* **140**, 21 (2016)
11. J. Striewe, C. Reuter, K.H. Sauerland, T. Tröster, *Thin-Walled Struct.* **123**, 501 (2018)
12. R.A. Eshkoo, S.A. Oshkovr, A.B. Sulong, R. Zulkifli, A.K. Ariffin, C.H. Azhari, *Compos. Pt B Eng.* **55**, 5 (2013)
13. H. Alshahrani, T.A. Sebaey, D.A. Hegazy, M.A.A. El-baky, *Polym. Adv. Tech.* **33**, 2921 (2022)
14. H. Alshahrani, T.A. Sebaey, M.M. Awd Allah, M.A. Abd El-baky, *Polym. Compos.* **43**(12), 8771–8783 (2022)
15. M.A. Abd El-baky, D.A. Hegazy, M.A. Hassan, *Appl. Compos. Mater.* **29**, 1195 (2022)
16. M.M. AwdAllah, A. Shaker, M.A. Hassan, M.A. Abd El-baky, *Polym. Compos.* **43**(11), 8322–8340 (2022)
17. A. Manoj, M. Bhuyan, S.R. Banik, M. RaviSankar, *Mater. Tod. Proceed.* **44**, 1375 (2021)
18. I. Gumperlein, E. Fischer, G. Dietrich-Gumperlein, S. Karrasch, D. Nowak, R.A. Jorres, R. Schierl, *Indoor Air* **28**, 611 (2018)
19. J. Grünewald, P. Parlevliet, V. Altstadt, J. Thermop. *Compos. Mater.* **30**, 437 (2016)
20. S.C. Ligon, R. Liska, J. Stampfl, M. Gurr, R. Mulhaupt, *Chem. Rev.* **117**, 10212 (2017)
21. M. Kucewicz, P. Baranowski, M. Stankiewicz, M. Konarzewski, P. Platek, J. Małachowski, *Thin-Walled Struct.* **145**, 106385 (2019)
22. M. Kucewicz, P. Baranowski, J. Małachowski, A. Poplawski, P. Platek, *Mater. Des.* **142**, 177 (2018)
23. B. Panda, M. Leite, B.B. Biswal, X. Niu, A. Garg, *Measurement* **116**, 495 (2018)
24. P. Zhang, D.J. Arceneaux, A. Khattab, *J. Appl. Polym. Sci.* **135**(12), 46018 (2018)
25. S. Qin, S. Cao, E. Yilmaz, J. Li, *Const. Build. Mater.* **307**, 124973 (2021)
26. G.D. Goh, S.J.C. Neo, V. Dikshit, W.Y. Yeong, J. Sand, *Struct. Mater.* **24**, 1206 (2021)
27. H. Li, Y. Hu, H. Huang, J. Chen, M. Zhao, B. Li, *Compos. Pt B Eng.* **215**, 108772 (2021)
28. C. Lubombo, M.A. Huneault, *Mater. Tod. Commun.* **17**, 214 (2018)
29. S. Ganeshkumar, S.D. Kumar, U. Magarajan, S. Rajkumar, B. Arulmurugan, S. Sharma, C. Li, R.A. Ilyas, M.F. Badran, *Materials* **15**(15), 5142 (2022)
30. Q. Ma, M.R.M. Rejab, A.P. Kumar, H. Fu, N.M. Kumar, J. Tang, *Proceed. Instit. Mech Eng. Pt C J. Mech. Eng. Sci.* **235**, 4254 (2020)
31. M.A. Quanjin, I.M. Sahat, M.R. Mat Rejab, S. Abu Hassan, B. Zhang, M.N.M. Merzuki, J. Reinforc. Plast. *Compos.* **38**, 1067 (2019)
32. M. Sadegh Ebrahimi, R. Hashemi, E. Etemadi, *J. Mater. Res. Tech.* **20**, 3616 (2022)
33. A. Tsouknidas, M. Pantazopoulos, I. Katsoulis, D. Fasnakis, S. Maropoulos, N. Michailidis, *Mater. Des.* **102**, 41 (2016)
34. M.S. Zahran, P. Xue, M.S. Esa, *Int. J. Crashworthiness* **22**, 294 (2016)
35. U. Morales, A. Esnaola, M. Irangi, L. Aretxabaleta, J. Aurrekoetxea, *Compos. Pt B Eng.* **217**, 108865 (2021)
36. K. Wang, Y. Liu, J. Wang, J. Xiang, S. Yao, Y. Peng, *Eng. Struct.* **254**, 11390 (2022)
37. Z. Wang, J. Deng, K. Liu, Y. Tao, *Thin-Walled Struct.* **171**, 108816 (2022)
38. Z. Jin, Z. Zhang, G.X. Gu, *Manufact. Lett.* **22**, 11 (2019)
39. M. Dawoud, I. Taha, S.J. Ebeid, *J. Manufact. Proc.* **21**, 39 (2016)
40. E. Linul, D. Pietras, T. Sadowski, L. Marşavina, D.K. Rajak, J. Kovacic, *Compos. Pt A Appl. Sci. Manufact.* **149**, 106516 (2021)
41. M.A. Abd El-baky, D.A. Hegazy, M.A. Hassan, M. Kamel, J. *Compos. Mater.* **56**, 1901 (2022)
42. K. Abd El-Aziz, D.A. Hegazy, M.A. Abd El-baky, *Fib. Polym.* **23**(8), 2284–2298 (2022)
43. M. Kathiresan, *Int. J. Crashworthiness* **27**, 1046 (2021)
44. M. Kathiresan, K. Manisekar, V. Manikandan, *Compos. Struct.* **108**, 584 (2014)
45. M. Kathiresan, K. Manisekar, V. Rajamohan, M.A. Güler, *Mech. Adv. Mater. Struct.* **29**(26), 5360–5377 (2022)
46. A. Baroutaji, A. Arjunan, G. Singh, J. Robinson, *Res. Eng.* **14**, 100424 (2022)
47. X. Zhao, G. Zhu, C. Zhou, Q. Yu, *Compos. Struct.* **222**, 110920 (2019)
48. E.M. Agaliotis, B.D. Ake-Concha, A. May-Pat, J.P. Morales-Arias, C. Bernal, A. Valadez-Gonzalez, P.J. Herrera-Franco, G. Proust, J.F. Koh-Dzul, J.G. Carrillo, E.A. Flores-Johnson, *Polymer* **14**(19), 3976 (2022)
49. F. Carrasco, P. Pagès, J. Gámez-Pérez, O.O. Santana, M.L. Maspoch, *Polym. Degrad. Stab.* **95**, 116 (2010)
50. Y. Liao, C. Liu, B. Coppola, G. Barra, L. Di Maio, L. Incarnato, K. Lafdi, *Polymers* **11**(9), 1487 (2019)
51. V. Figueroa-Velarde, T. Diaz-Vidal, E.O. Cisneros-López, J.R. Robledo-Ortiz, E.J. López-Naranjo, P. Ortega-Gudiño, L.C. Rosales-Rivera, *Materials* **14**(11), 3111 (2021)
52. H. Liu, H. He, X. Peng, B. Huang, J. Li, *Poly. Advanc. Tech.* **30**, 910 (2019)
53. H. Alshahrani, T.A. Sebaey, M.M. Awd Allah, M.A. Abd El-baky, *J. Compos. Mater.* **56**(27), 4195–4218 (2022)
54. E.F. Abdewi, S. Sulaiman, A.M.S. Hamouda, E. Mahdi, *Thin-Walled Struct.* **46**, 320 (2008)
55. M.A. Guler, M.E. Cerit, B. Bayram, B. Gerçek, E. Karakaya, *Int. J. Crashworthiness* **15**, 377 (2010)
56. H. Alshahrani, T.A. Sebaey, M.M. Awd Allah, M.A. Abd El-baky, *J. Compos. Mater.* **57**, 1579 (2023)
57. M.I.A. El Aal, M.M. Awd Allah, M.A. Abd El-baky, *Polym. Compos* **44**(5), 2910–24 (2023)
58. A. Mamalis, D. Manolacos, G. Demosthenous, E. Ioannidis, *Compos. Pt B Eng.* **28**, 439 (1997)
59. M. A. A. El-baky, M. M. A. Allah, M. Kamel, and W. Abdel-Aziem, *Fib. Polym.*, (2023).